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ORIGINAL ARTICLE

Unimodality of the Independence Polynomial of $G \circ (K_p \cup K_q)$

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Abstract
In this paper, our research primarily focuses on f-symmetric and unimodal properties of independence polynomials $I(G \circ (K_p \cup K_q); x)$. More precisely, we prove that the independence polynomial $I(G \circ (K_p \cup K_q); x)$ is unimodal for every graph G, whenever the positive integers p and q are large enough. In addition, we establish several inequalities involving the coefficients of $I(G \circ (K_p \cup K_q); x)$ for arbitrary p and q.

Keywords Independence polynomial · Unimodality · f-symmetric polynomial

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1 Introduction

Throughout this paper $G = (V, E)$ is a finite, undirected, loopless graph without multiple edges, with vertex set $V = V(G)$ of cardinality $|V(G)| = n(G)$, and edge set $E = E(G)$ of size $|E(G)| = m(G)$. An edge $e \in E(G)$ connecting two vertices x and y will also be written as xy. In this case, it is said that x and y are *adjacent*. The neighborhood of a vertex x in G is the set $N_G(x) = \{y \in V(G) \mid xy \in E(G)\}$ and the closed neighborhood of x in G is the set $N_G[x] = \{x\} \cup N_G(x)$. For a subset S of $V(G)$, let $G[S]$ be the induced subgraph of G on the vertex set S, and we use $G - S$ to denote $G[V(G) - S]$. If $S = \{x\}$, we write

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$G - x$ instead of $G - \{x\}$. A set $S \subseteq V(G)$ is *independent* if no two vertices from S are adjacent, and by $\mathrm{Ind}(G)$ we mean the family of all the independent sets of G. An independent set of G is *maximal* if it cannot be extended to a larger independent set. An independent set of maximum size is a *maximum independent set* of G, and $\alpha(G) = \max\{|S| : S \in \mathrm{Ind}(G)\}$.

The disjoint union G_1 and G_2, denoted $G_1 \cup G_2$, is the graph with the vertex set $V(G_1) \cup V(G_2)$ and the edge set $E(G_1) \cup E(G_2)$. In particular, mG denotes the disjoint union of $m > 1$ copies of the graph G. The Zykov sum of G_1 and G_2, denoted $G_1 + G_2$, is the graph with $V(G_1 + G_2) = V(G_1) \cup V(G_2)$ and $E(G_1 + G_2) = E(G_1) \cup E(G_2) \cup \{v_1v_2 : v_1 \in V(G_1), v_2 \in V(G_2)\}$.

If s_k equals the number of independent sets of cardinality k in the graph G, then the *independence polynomial* of G is defined as follows:

$$I(G; x) = \sum_{k=0}^{\alpha(G)} s_k x^k = s_0 + s_1 x + \cdots + s_{\alpha(G)} x^{\alpha(G)}.$$

The independence polynomial, first introduced by Gutman and Harary in 1983 [12], has been extensively discussed in a substantial number of publications. Recall that a polynomial $P(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ is:

- *unimodal* if there exists an index $0 \le k \le n$, called the *mode* of the $P(x)$, such that

$$a_0 \le a_1 \le \cdots \le a_k \ge a_{k+1} \ge \cdots \ge a_n;$$

- *log-concave* if $a_i^2 \ge a_{i-1} \cdot a_{i+1}$ holds for all $1 \le i \le n - 1$;
- *symmetric* if $a_{n-i} = a_i$ for all $0 \le i \le \lfloor \frac{n}{2} \rfloor$;
- *f-symmetric* if $a_{n-i} = f(i) \cdot a_i$ for all $0 \le i \le \lfloor \frac{n}{2} \rfloor$ [20].

It is known that $I(G; x)$ is unimodal whenever it is log-concave. It is worth reminding that 1-symmetric polynomials are also known as self-reciprocal or palindromic polynomials.

The problem of unimodality has been thoroughly investigated across multiple areas, including algebra, combinatorics, and geometry (see [1, 3, 4, 6–8, 13, 18, 19, 21, 22, 24, 25, 30] and their references). The independence polynomial of an arbitrary graph G is not necessarily unimodal or symmetric. For instance, the polynomial $I(K_{127} + 3K_7; x) = 1 + 148x + 147x^2 + 343x^3$ is neither unimodal nor symmetric.

In [1], Alavi, Malde, Schwenk and Erdös proved that for any permutation π of $\{1, 2, \ldots, \alpha\}$ there is a graph G with $\alpha(G) = \alpha$ such that $s_{\pi(1)} < s_{\pi(2)} < \ldots < s_{\pi(\alpha)}$. Nevertheless, for trees, they proposed the following conjecture, which is still open.

Conjecture 1.1 [1] The independence polynomial of a tree is unimodal.

Recall that a graph G is *well-covered* if all its maximal independent sets are of equal size [29]. It has been shown that the independence polynomial of well-covered graphs is not unimodal in general [23]. Some more findings related to well-covered trees may be found in [16] and [17]. Despite these results, Conjecture 1.1 remains unresolved. Actually, the independence polynomial of well-covered graphs may sufficiently vary from unimodality [23, 28]. In this context, it is worth mentioning the following conjecture regarding the partial unimodality of the coefficients of the independence polynomial.

Conjecture 1.2 *(Roller–Coaster Conjecture)* [28] For every permutation σ of the set $\{\lceil \frac{\alpha}{2} \rceil, \ldots, \alpha\}$, there is a well-covered graph G with $\alpha(G) = \alpha$ such that the coefficients of $I(G; x)$ satisfy $s_{\sigma(\lceil \frac{\alpha}{2} \rceil)} < s_{\sigma(\lceil \frac{\alpha}{2} \rceil)+1} < \cdots < s_{\sigma(\alpha)}$.

The Roller–Coaster Conjecture has been verified for well-covered graphs G having $\alpha(G) \leq 7$ [28], and later for $\alpha(G) \leq 11$ [27]. This conjecture has been validated in [9], and further strengthened for several classes of graphs [23]. Therefore, it is important to identify specific classes of graphs whose independence polynomials exhibit partial unimodality.

In this work, we focus on the corona graph $G \circ \mathcal{H}$. Let $\mathcal{H} = \{H_v : v \in V(G)\}$ be a family of non-empty graphs indexed by the vertex set of the graph G. The *corona* $G \circ \mathcal{H}$ is defined as the disjoint union of G and H_v for each $v \in V(G)$, with additional edges connecting each vertex v to all vertices of H_v [10]. When $H_v = H$ for all $v \in V(G)$, the corona graph $G \circ \mathcal{H}$ is denoted $G \circ H$. Furthermore, the second and third authors demonstrated in [21] that if $\alpha(H) = 2$, then $I(G \circ H; x)$ is f-symmetric with $f(i) = \left(\frac{n(H)(n(H)-1)}{2} - m(H)\right)^{n(G)-i}$, and, consequently, $I(G \circ (K_r - e); x)$ is both symmetric and unimodal for all complete graphs K_r with $r \geq 2$ and $e \in E(K_r)$. This brings up the question of what sufficient conditions ensure unimodality of $I(G \circ H; x)$, when $\alpha(H) = 2$. In this paper, we concentrate on the unimodality of the independence polynomial for $G \circ (K_p \cup K_q)$, where p and q are positive integers. Specifically, our first main result reads as follows.

Theorem 1 *Let G be a graph of order n. If $\frac{p \cdot q+1}{p+q+1} \geq n$, then $I(G \circ (K_p \cup K_q); x)$ is f-symmetric and unimodal, where $f(i) = (p \cdot q)^{n-i}$ for all $0 \leq i \leq n$.*

Additionally, Lemma 5.1 establishes relationships between the coefficients of $I(G \circ (K_p \cup K_q); x) = \sum_{m=0}^{2n} t_m x^m$ and $I(G; x) = \sum_{i=0}^{\alpha(G)} s_i x^i$, specifically:

$$t_m = \sum_{j=0}^{m} \sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k} s_j$$

for all $0 \leq m \leq 2n$.

The second main finding is as follows.

Theorem 2 *Let p, q be positive integers, $n = n(G)$, and $I(G \circ (K_p \cup K_q); x) = \sum_{i=0}^{2n} t_i x^i$. Then the following assertions hold true:*

1. *$t_0 \leq t_1 \leq \cdots \leq t_r \leq t_{r+1}$, where $r = \max\{\lfloor \frac{np-1}{p+1} \rfloor, \lfloor \frac{nq-1}{q+1} \rfloor\}$. In particular, if $\min\{p, q\} \geq n$, then $t_0 \leq t_1 \leq \cdots \leq t_n$.*
2. *$t_s \geq t_{s+1} \geq \cdots \geq t_{2n}$, whenever $s = \left\lceil \frac{n(p+q+1)-1}{\min\{p,q,\frac{p+q-1}{2}\}+2} \right\rceil$ and $s \leq 2n-1$.*

Consequently, Corollary 5.6 demonstrates a significant improvement in the increasing and decreasing trends of the coefficients of $I(G \circ 2K_p; x)$. In particular, the unimodality of $I(G \circ 2K_1; x)$, first established in [26], follows from this result.

This paper is organized as follows. Section 2 reviews preliminaries related to f-symmetric and unimodal properties. In Section 3, we present the proof of our first main theorem. In Section 4, we conduct an investigation into the λ-quasi-regularizable property, while Section 5 provides the proof of our second key finding. Finally, in Conclusion, we suggest potential directions for future research.

2 Preliminaries

These following formulas are helpful tools to compute the independence polynomials.

Lemma 2.1 [11, 12, 14] *Let G and H be two disjoint graphs. The following assertions are true:*

1. $I(G \cup H; x) = I(G; x) \cdot I(H; x)$;
2. $I(G; x) = I(G - v; x) + x \cdot I(G - N[v]; x)$ *holds for every* $v \in V(G)$.

It is known that the product of two unimodal polynomials is not necessarily unimodal, for example let $G = K_{95} + 3K_7$, then $I(G; x) = 1 + 116x + 147x^2 + 343x^3$, while $I(2G; x) = 1 + 232x + 13750x^2 + 34790x^3 + 101185x^4 + \mathbf{100842}x^5 + 117649x^6$.

Lemma 2.2 [15] *If $P(x)$ is log-concave and $Q(x)$ is unimodal, then $P(x) \cdot Q(x)$ is unimodal, while the product of two log-concave polynomials is log-concave.*

The sum of two unimodal and symmetric polynomials is not necessarily symmetric and/or unimodal. Naturally, one might question whether the sum and product of two f-symmetric and unimodal polynomials are also f-symmetric and unimodal. In case that both $P(x)$ and $Q(x)$ are f-symmetric with $f(i) = (p \cdot q)^{n-i}$ we obtain the following.

Lemma 2.3 *Let p and q be two positive integers. Let $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ be a function such that $f(i) = (p \cdot q)^{n-i}$ for all $1 \le i \le n$. Assume $P(x) = \sum_{i=0}^{2n} a_i x^i$ is an f-symmetric and unimodal polynomial with positive coefficients. If $p \cdot q \ge \frac{a_1}{a_0}$, then the mode of $P(x)$ is $2n$.*

Proof Since $P(x)$ is f-symmetric, we have

$$\frac{a_{2n}}{a_{2n-1}} = \frac{(p \cdot q)^{2n} \cdot a_0}{(p \cdot q)^{2n-1} \cdot a_1} = \frac{p \cdot q \cdot a_0}{a_1} \ge 1.$$

Thus $a_{2n} \ge a_{2n-1}$. On the other hand, $P(x)$ is unimodal and, consequently, the mode of $P(x)$ is $2n$, as required. □

Lemma 2.4 *Let $f : \mathbb{Z}_{>0} \to \mathbb{Z}_{>0}$ be a function. For $r \ge 2$, let $P(x)$ and $Q(x)$ be two polynomials of degree r and $r-1$, respectively. Assume that $P(0) \ne 0$ and $Q(0) = 0$.*

1. *If both $P(x)$ and $Q(x)$ are f-symmetric, then so is $P(x) + Q(x)$.*
2. *If both $P(x)$ and $Q(x)$ are unimodal with the same mode $1 \le k \le r-1$, then $P(x)+Q(x)$ is also unimodal with mode at k.*
3. *If $P(x)$ is unimodal with mode at r and $Q(x)$ is unimodal with mode at $r-1$, then $P(x) + Q(x)$ is also unimodal.*

Proof Let $P(x) = \sum_{i=0}^{r} a_i x^i$ and $Q(x) = \sum_{i=0}^{r-1} b_i x^i$. By the hypothesis, $Q(0) = 0$, so $b_0 = 0$. Then

$$P(x) + Q(x) = a_0 + \sum_{i=1}^{r-1} (a_i + b_i) \cdot x^i + a_r \cdot x^r.$$

1. We have $a_r = f(0) \cdot a_0$ and $a_{r-i} + b_{r-i} = f(i) \cdot a_i + f(i) \cdot b_i = f(i) \cdot (a_i + b_i)$ for all $1 \le i \le r-1$. Thus, $P(x) + Q(x)$ is f-symmetric.
2. By the hypothesis, since both $P(x)$ and $Q(x)$ are unimodal and have the same mode k, we obtain

$$a_0 \le a_1 \le \cdots \le a_k \ge \cdots \ge a_{r-1} \ge a_r, \quad \text{and}$$
$$0 = b_0 \le b_1 \le \cdots \le b_k \ge \cdots \ge b_{r-1} \ge b_r = 0.$$

It implies that

$$a_0 + b_0 \le a_1 + b_1 \le \cdots \le a_k + b_k \ge \cdots \ge a_{r-1} + b_{r-1} \ge a_r + b_r,$$

as required.

3. By the assumption,

$$a_0 + b_0 \le a_1 + b_1 \le \cdots \le a_{r-1} + b_{r-1}.$$

If $a_{r-1} + b_{r-1} \le a_r$, then $P(x) + Q(x)$ is unimodal with mode at r. Otherwise, $P(x) + Q(x)$ is unimodal with mode at $r - 1$. □

Lemma 2.5 *Let p, q be two positive integers, and $P(x) = \sum_{i=0}^{2n} a_i \cdot x^i$ be an f-symmetric polynomial of degree $2n$, where $f(i) = (p \cdot q)^{n-i}$ for all $1 \le i \le n$. Then*

1. *$(1 + px) \cdot (1 + qx) \cdot P(x)$ is also f-symmetric, where $f(i) = (p \cdot q)^{n-i+1}$ for all $0 \le i \le n + 1$.*
2. *Moreover, if $P(x)$ is unimodal and $p \cdot q - p - q \ge \frac{a_1}{a_0}$, then $(1 + px) \cdot (1 + qx) \cdot P(x)$ is unimodal with mode $2n + 2$.*

Proof Let $Q(x) = (1 + px) \cdot (1 + qx) \cdot P(x) = \sum_{i=0}^{2n+2} b_i \cdot x^i$. Then

$$Q(x) = a_0 + [a_1 + (p + q) \cdot a_0] \cdot x + \sum_{i=2}^{2n} [a_i + (p + q) \cdot a_{i-1} + p \cdot q \cdot a_{i-2}] \cdot x^i$$
$$+ [(p + q) \cdot a_{2n} + p \cdot q \cdot a_{2n-1}] \cdot x^{2n+1} + p \cdot q \cdot a_{2n} \cdot x^{2n+2}.$$

1. By the hypothesis, we have $a_{2n-i} = (p \cdot q)^{n-i} \cdot a_i$ for all $0 \le i \le n$. Then

$$\begin{aligned} b_{2n+2} &= p \cdot q \cdot a_{2n} = (p \cdot q)^{n+1} \cdot a_0 = (p \cdot q)^{n+1} \cdot b_0, \\ b_{2n+1} &= (p + q) \cdot a_{2n} + p \cdot q \cdot a_{2n-1} = (p + q) \cdot (p \cdot q)^n \cdot a_0 + (p \cdot q) \cdot (p \cdot q)^{n-1} a_1 \\ &= (p \cdot q)^n \cdot [(p + q) \cdot a_0 + a_1] = (p \cdot q)^n \cdot b_1, \end{aligned}$$

and

$$\begin{aligned} b_{2n+2-i} &= a_{2n+2-i} + (p + q) \cdot a_{2n+1-i} + p \cdot q \cdot a_{2n-i} \\ &= (p \cdot q)^{n-(i-2)} \cdot a_{i-2} + (p + q) \cdot (p \cdot q)^{n-(i-1)} \cdot a_{i-1} + p \cdot q \cdot (p \cdot q)^{n-i} \cdot a_i \\ &= (p \cdot q)^{n-i+1} \cdot [p \cdot q \cdot a_{i-2} + (p + q) \cdot a_{i-1} + a_i] \\ &= (p \cdot q)^{n-i+1} \cdot b_i, \end{aligned}$$

for all $2 \le i \le n + 1$.

2. By Lemma 2.2, $Q(x)$ is unimodal and

$$\frac{b_1}{b_0} = \frac{a_1 + (p + q) \cdot a_0}{a_0} = \frac{a_1}{a_0} + p + q \le p \cdot q,$$

while the mode of $P(x)$ is $2n$ in accordance with Lemma 2.3. □

Lemma 2.6 *Let p, q, k be three positive integers, and $P(x) = \sum_{i=0}^{2n} a_i x^i$ be an f-symmetric polynomial of degree $2n$, where $f(i) = (p \cdot q)^{n-i}$ for all $1 \le i \le n$. Then*

1. *$(1 + px)^k \cdot (1 + qx)^k \cdot P(x)$ is also f-symmetric, where $f(i) = (p \cdot q)^{n-i+k}$ for all $1 \le i \le n + k$.*
2. *Moreover, if $P(x)$ is unimodal and $p \cdot q - k \cdot (p + q) \ge \frac{a_1}{a_0}$, then $(1+px)^k \cdot (1+qx)^k \cdot P(x)$ is unimodal with mode $2n + 2k$.*

Proof Let $(1 + px)^k \cdot (1 + qx)^k \cdot P(x) = \sum_{i=0}^{2n+2k} c_i \cdot x^i$. Note that $c_0 = a_0$ and $c_1 = a_1 + k \cdot (p + q) \cdot a_0$. Using Lemma 2.5, one can complete the proof by the induction on k. □

3 The Proof of Theorem 1

The aim of this section is to show that the independence polynomial of $G \circ (K_p \cup K_q)$ is unimodal under some conditions on p, q, and $n(G)$.

Lemma 3.1 [20] *The polynomial $I(G \circ (K_p \cup K_q); x)$ is f-symmetric of degree $2n(G)$, where $f(i) = (p \cdot q)^{n(G)-i}$ for all $1 \le i \le n(G)$.*

Proof of Theorem 1 Denote $L = G \circ (K_p \cup K_q)$. By Lemma 3.1, we know that $I(L; x)$ is f-symmetric with $f(i) = (p \cdot q)^{n(G)-i}$ for all $0 \le i \le n(G)$. Now, we prove that $I(L; x)$ is unimodal, by induction on $n = n(G)$.

If $E(G) = \emptyset$, then

$$I(nK_1 \circ (K_p \cup K_q); x) = [(1 + px) \cdot (1 + qx) + x]^n = \left[1 + (p + q + 1) \cdot x + p \cdot q \cdot x^2\right]^n$$

is log-concave (and, consequently, unimodal), in accordance with Lemma 2.2 and the fact that $1 + (p + q + 1) \cdot x + p \cdot q \cdot x^2$ is log-concave.

Suppose $n \ge 2$ and $E(G) \ne \emptyset$. Let $v \in V(G)$. Assume $|N_G(v)| = k \ge 1$. By Lemma 2.1(2), $I(L; x) = I(L - v; x) + x \cdot I(L - N_L[v]; x)$. Since,

$$L - v = ((G - v) \circ (K_p \cup K_q)) \cup (K_p \cup K_q) \quad \text{and}$$
$$L - N_L[v] = ((G - N_G[v]) \circ (K_p \cup K_q)) \cup k(K_p \cup K_q),$$

by Lemma 2.1(1), we obtain

$$I(L - v; x) = (1 + px) \cdot (1 + qx) \cdot I((G - v) \circ (K_p \cup K_q); x),$$
$$I(L - N_L[v]; x) = (1 + px)^k \cdot (1 + qx)^k \cdot I((G - N_G[v]) \circ (K_p \cup K_q); x).$$

By the induction hypothesis, both $I((G-v) \circ (K_p \cup K_q); x)$ and $I((G-N_G[v]) \circ (K_p \cup K_q); x)$ are unimodal. Furthermore, we prove the following two claims.

Claim 1 $I(L - v; x)$ is f-symmetric and unimodal with mode $2n$, where $f(i) = (p \cdot q)^{n-i}$ for all $0 \le i \le n$.

Proof Indeed, Lemma 3.1 implies that $I((G-v) \circ (K_p \cup K_q); x)$ is f-symmetric with $f(i) = (p \cdot q)^{n-1-i}$ for all $0 \le i \le n-1$. Thus, by Lemma 2.5(1), $I(L - v; x)$ is f-symmetric with $f(i) = (p \cdot q)^{n-i}$ for all $0 \le i \le n$. Moreover, let $I((G-v) \circ (K_p \cup K_q); x) = \sum_{i=0}^{2n-2} b_i \cdot x^i$. Thus $p \cdot q - p - q \ge \frac{b_1}{b_0}$, because $\frac{p \cdot q + 1}{p+q+1} \ge n$. Now, the claim follows from Lemma 2.5(2).□

Claim 2 $I(L - N_L[v]; x)$ is f-symmetric and unimodal with mode $2n - 2$, where $f(i) = (p \cdot q)^{n-i-1}$ for all $0 \le i \le n - 1$.

Proof Indeed, Lemma 3.1 implies that $I((G - N_G[v]) \circ (K_p \cup K_q); x)$ is f-symmetric with $f(i) = (p \cdot q)^{n-i-k-1}$ for all $0 \le i \le n - k - 1$. Thus, by Lemma 2.6(1), $I(L - N_L[v]; x)$ is also f-symmetric with $f(i) = (p \cdot q)^{n-i-1}$ for all $0 \le i \le n - 1$. Now, let

$$I((G - N_G[v]) \circ (K_p \cup K_q); x) = \sum_{i=0}^{2n-2k-2} c_i \cdot x^i.$$

Clearly, $c_0 = 1$ and $c_1 = (n - k - 1) \cdot (p + q + 1)$. Since $\frac{p \cdot q + 1}{p+q+1} \ge n$, we get

$$\begin{aligned} p \cdot q - k \cdot (p + 1) &\ge n(p + q + 1) - 1 - k \cdot (p + q) \\ &= (n - k - 1) \cdot (p + q + 1) + (p + q + k) \\ &\ge (n - k - 1) \cdot (p + q + 1) = \frac{c_1}{c_0}. \end{aligned}$$

Applying Lemma 2.6(2), we infer that $I(L - N_L[v]; x)$ is unimodal with mode $2n-2$, as claimed. □

Finally, according to Lemma 2.4(3), we obtain that $I(G \circ (K_p \cup K_q); x)$ is unimodal. □

4 Quasi-regularizable Graphs

For $\lambda > 0$, a graph H is called λ-*quasi-regularizable* if $\lambda \cdot |S| \leq |N_H(S)|$ for every independent set S of H [23]. For two positive integers $\lambda \geq \lambda'$, by this definition, we can deduce that if H is λ-quasi-regularizable, then H has no isolated vertices and moreover it is also λ'-quasi-regularizable. In the case $\lambda = 1$, H is called a *quasi-regularizable* graph. It is known that well-covered graphs without isolated vertices are quasi-regularizable [5]. The *vertex expansion ratio* of an independent set S in the graph H is defined as $\lambda_H(S) = \frac{|N_H(S)|}{|S|}$, where in the case S is an empty set we consider $\lambda_H(S)$ to be infinite. [2]. The *vertex expansion ratio* of the graph H is defined as $\lambda(H) = \min_{S \in \mathrm{Ind}(H)} \lambda_H(S)$. Clearly, if the graph H has no isolated vertices, then it is λ-quasi-regularizable for every $\lambda \leq \lambda(H)$. Moreover,

$$\lambda(H) = \min_{S \in \mathrm{Ind}(H)} \lambda_H(S) = \max_{k \cdot |S| \leq |N_H(S)|, \forall S \in \mathrm{Ind}(H)} k.$$

Lemma 4.1 [23] *For $\lambda > 0$, let H be a λ-quasi-regularizable graph and $I(H; x) = \sum_{k=0}^{\alpha(H)} s_k \cdot x^k$. Then the following assertions are true:*

1. $(k+1) \cdot s_{k+1} \leq (n(H) - (\lambda+1) \cdot k) \cdot s_k$, $0 \leq k < \alpha(H)$;
2. *if $n(H) \leq (\lambda+2) \cdot \alpha(H) - (\lambda+1)$ (i.e., $\lceil \frac{n(H)-1}{\lambda+2} \rceil \leq \alpha(H) - 1$), then $s_r \geq s_{r+1} \geq \cdots \geq s_{\alpha(H)}$, where $r = \lceil \frac{n(H)-1}{\lambda+2} \rceil$.*

Lemma 4.2 *Let p, q be two positive integers. Suppose that r, s, t are three non-negative integers such that $(s, r, t) \neq (0, 0, 0)$. Then*

$$\frac{pr + qs + (p+q-1)t}{r+s+2t} \geq \min\left\{p, q, \frac{p+q-1}{2}\right\}.$$

Proof We divide the proof into the following three cases:
Case 1. $q \geq p+1$. We can see that

$$\begin{aligned} pr + qs + (p+q-1)t - p(r+s+2t) &= qs - ps - pt - t + qt \\ &= s(q-p) + t(q-p-1) \geq 0. \end{aligned}$$

Therefore,

$$\frac{pr + qs + (p+q-1)t}{r+s+2t} \geq p.$$

Case 2. $p \geq q+1$. In this case, we always have $r(p-q) + t(p-q-1) \geq 0$. This implies that

$$\begin{aligned} pr + qs + (p+q-1)t - q(r+s+2t) &= pr - t - qr + pt - qt \\ &= r(p-q) + t(p-q-1) \geq 0. \end{aligned}$$

Therefore,

$$\frac{pr + qs + (p+q-1)t}{r+s+2t} \geq q.$$

Case 3. $p = q$. In this case, the inequality to be demonstrated is

$$\frac{pr + ps + (2p-1)t}{r+s+2t} \geq \frac{2p-1}{2},$$

which is true, because $2(pr + ps + (2p-1)t) - (2p-1)(r+s+2t) = r + s \geq 0$. □

Proposition 4.3 *The graph* $G \circ (K_p \cup K_q)$ *is* $\min\{p, q, \frac{p+q-1}{2}\}$*-quasi-regularizable for any two positive integers* p *and* q.

Proof Let $H = G \circ (K_p \cup K_q)$. Let $n = n(G)$ and $\lambda = \lambda(H)$. Our goal is to demonstrate that

$$\lambda = \min\left\{p, q, \frac{p+q-1}{2}\right\}.$$

Indeed, first, for $a \in V(K_p)$ and $b \in V(K_q)$, $|N_H(a)| = p$ and $|N_H(b)| = q$. It follows that $p = \lambda_H(\{a\}) \geq \lambda$ and $q = \lambda_H(\{b\}) \geq \lambda$. Next, take $v \in V(G)$. There are $a \in V(K_p)$ and $b \in V(K_q)$ such that v is adjacent to both a and b. Note that $|N_H(\{a, b\})| = p + q - 1$. Then $\frac{p+q-1}{2} = \lambda_H(\{a, b\}) \geq \lambda$. Therefore,

$$\lambda \leq \min\left\{p, q, \frac{p+q-1}{2}\right\}. \tag{1}$$

In order to prove the opposite inequality, we need to show that

$$\lambda_H(S) \geq \min\left\{p, q, \frac{p+q-1}{2}\right\}$$

for all $S \in \mathrm{Ind}(H)$. Now, let S be an independent set of H. We partition the set S into two disjoint subsets $S_1 \subseteq V(G)$ and $S_2 \subseteq V(H) - V(G)$. Since each $v \in S_1$ has at least $p + q$ neighbors in $nK_p \cup nK_q$, it yields $|N_H(S_1)| \geq (p+q) \cdot |S_1|$. If $S_2 = \emptyset$, then

$$\lambda_H(S) = \lambda_H(S_1) \geq p + q \geq \min\left\{p, q, \frac{p+q-1}{2}\right\},$$

as required.

Now we consider the case with $S_2 \neq \emptyset$. Based on the structure of graph H, we may assume that

$$S_2 = \{a_1, \ldots, a_r\} \cup \{b_1, \ldots, b_s\} \cup \{c_1, d_1, \ldots, c_t, d_t\},$$

such that

- $a_i \in V(nK_p)$ is adjacent to $v \in V(G)$ and $N_H(v) \cap S_2 = \{a_i\}$ for all $1 \leq i \leq r$;
- $b_i \in V(nK_q)$ is adjacent to $v \in V(G)$ and $N_H(v) \cap S_2 = \{b_i\}$ for all $1 \leq i \leq s$;
- $c_i \in V(nK_p)$ and $d_i \in V(nK_q)$ are adjacent to a vertex $v_i \in V(G)$ for all $1 \leq i \leq t$.

Observe that $(s, r, t) \neq (0, 0, 0)$ since $S_2 \neq \emptyset$. Additionally, $|S_2| = r + s + 2t$ and $|N_H(S_2)| = pr + qs + (p+q-1)t$. Thus $|N_H(S_2)| \leq (p+q) \cdot |S_2|$. Consequently,

$$\begin{aligned}
\lambda_H(S) = \lambda_H(S_1 \cup S_2) &= \frac{|N_H(S_1)| + |N_H(S_2)|}{|S_1| + |S_2|} \\
&\geq \frac{(p+q) \cdot |S_1| + |N_H(S_2)|}{|S_1| + |S_2|} \\
&\geq \lambda_H(S_2) = \frac{pr + qs + (p+q-1)t}{r+s+2t}.
\end{aligned}$$

By Lemma 4.2, $\lambda_H(S) \geq \min\left\{p, q, \frac{p+q-1}{2}\right\}$. Therefore,

$$\lambda = \lambda(H) = \min_{S \in \mathrm{Ind}(H)} \lambda_H(S) \geq \min\left\{p, q, \frac{p+q-1}{2}\right\}.$$

Together with the inequality (1), we infer that

$$\lambda = \min\left\{p, q, \frac{p+q-1}{2}\right\}.$$

In summary, $H = G \circ (K_p \cup K_q)$ is $\min\left\{p, q, \frac{p+q-1}{2}\right\}$-quasi-regularizable. □

Proposition 4.3 immediately implies the following.

Corollary 4.4 *The graph $G \circ 2K_p$ is $\left(p - \frac{1}{2}\right)$-quasi-regularizable for every positive integer p.*

Example 4.5 The graph $H = C_5 \circ 2K_2$, shown in Fig. 1, is $\frac{3}{2}$-quasi-regularizable but not 2-regularizable.

Indeed, let $S = \{a_1, a_2, \ldots, a_5\} \cup \{c_1, c_2, \ldots, c_5\}$. It is clear that S is a maximum independent set in H. Then $N_H(S) = \{b_1, \ldots, b_5\} \cup \{d_1, \ldots, d_5\} \cup \{x_1, \ldots, x_5\}$. Hence, $|S| = 10$ and $|N_H(S)| = 15$. This concludes that H is not 2-quasi-regularizable.

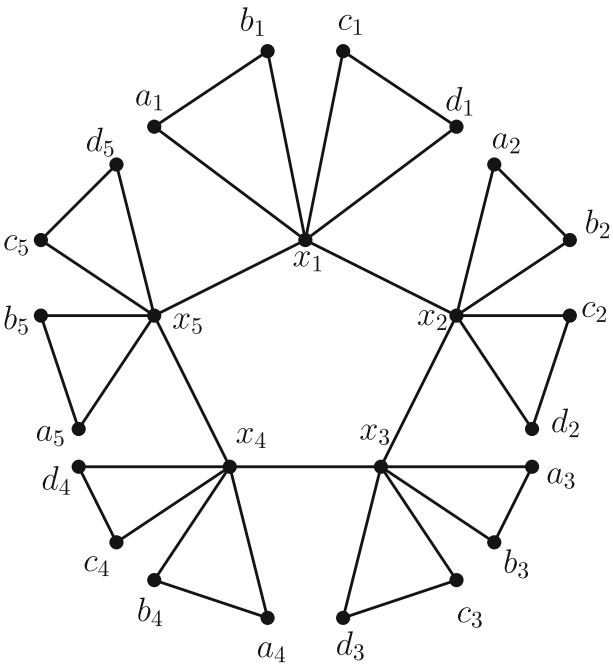


Fig. 1 The graph $C_5 \circ 2K_2$

5 The Proof of Theorem 2

Throughout this section, let G be a graph of the order n, $I(G;x) = \sum_{k=0}^{\alpha(G)} s_j \cdot x^j$, p and q be two positive integers, and $I(G \circ (K_p \cup K_q);x) = \sum_{m=0}^{2n} t_m x^m$. Clearly, $2n = \alpha(G \circ (K_p \cup K_q))$.

In this section, by relating the coefficient t_m with s_j, we further investigate the partial unimodality problem of $I(G \circ (K_p \cup K_q);x)$.

Lemma 5.1 *For $0 \le m \le 2n$, we have*

$$t_m = \sum_{j=0}^{m} \sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k} s_j.$$

Proof Let S be an independent set of size m in $G \circ (K_p \cup K_q)$. Assume that $S = S_1 \cup S_2$ with $S_1 \subseteq V(G)$ and $S_2 \subseteq V(nK_p \cup nK_q)$. Let $|S_1| = j$ and $|S_2| = m - j$. Since S_1 is an independent set of G, it yields $0 \le j \le \alpha(G)$. Then we have s_j choices for S_1.

Because of the independence of S, we deduce that

$$S_2 \subseteq V(nK_p \cup nK_q) - \bigcup_{v_i \in S_1} (V(K_{p_i}) \cup V(K_{q_i})).$$

Notice that

$$\begin{aligned} &V(nK_p \cup nK_q) - \bigcup_{v_i \in S_1} (V(K_{p_i}) \cup V(K_{q_i})) \\ &= \left(V(K_{p_1}) \cup V(K_{q_1})\right) \cup \cdots \cup \left(V(K_{p_{n-j}}) \cup V(K_{q_{n-j}})\right), \end{aligned}$$

where $p_k = p$ and $q_k = q$ for all $1 \le k \le n - j$.

Assume that $S_2 = A \cup B$, where $A \subseteq \bigcup_{i=1}^{n-j} V(K_{p_i})$ and $B \subseteq \bigcup_{i=1}^{n-j} V(K_{q_i})$. If $|A| = k$, then $|B| = m - j - k$. Note that $0 \le k \le m - j$.

Let us enumerate all the sets satisfying the above constraints. In order to define the set A, we choose k sets from $n - j$ sets $V(K_{p_1}), \ldots, V(K_{p_{n-j}})$, and in each chosen set, we select an element. The set of all these elements forms A. Hence, the number of choices for A is $\binom{n-j}{k} p^k$. Similarly, the number of choices for B is $\binom{n-j}{m-j-k} q^{m-j-k}$. Since $A \cap B = \emptyset$, the number of choices for $S_2 = A \cup B$ is

$$\sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k}.$$

Hence, we have

$$\sum_{j=0}^{m} \left(\sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k} \right) s_j$$

choices for S and so

$$t_m = \sum_{j=0}^{m} \sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k} s_j,$$

as claimed. □

Remark 5.2 Let $\alpha = \alpha(G)$. Note that $s_j = 0$ whenever $j > \alpha$. Thus, if $m > \alpha$, then

$$t_m = \sum_{j=0}^{\alpha} \sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k} s_j.$$

Example 5.3 1. $I(K_3 \circ 2K_1; x) = 1 + 9x + 27x^2 + \mathbf{38}x^3 + 27x^4 + 9x^5 + x^6$.
2. $I(K_{1,3} \circ (K_8 \cup K_{10}); x) = 1 + 76x + 2483x^2 + 45565x^3 + 513774x^4 + 3645200x^5 + 15891200x^6 + 38912000x^7 + \mathbf{40960000}x^8$.

Furthermore, we make use of the following well-known formula.

Lemma 5.4 *Let $m, n, i, j, k \in \mathbb{N}$, $0 \le i \le n$, and $0 \le j \le m$. Then the following equality holds true:*

$$\sum_{\substack{i+j=k,\\ i,j \ge 0}} \binom{n}{i} \binom{m}{j} = \binom{n+m}{k}.$$

Corollary 5.5 *If $I(G \circ 2K_p; x) = \sum_{m=0}^{2n} t_m x^m$, then*

$$t_m = \sum_{j=0}^{m} \binom{2n-2j}{m-j} p^{m-j} s_j.$$

Proof Applying Lemma 5.1, we have

$$t_m = \sum_{j=0}^{m} \left[\sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} \right] p^{m-j} s_j.$$

Hence, by Lemma 5.4, we obtain that

$$\sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} = \sum_{\substack{s+t=m-j,\\ s,t \ge 0}} \binom{n-j}{s} \binom{n-j}{t} = \binom{2n-2j}{m-j},$$

as claimed. □

Proof of Theorem 2 By Lemma 5.1, we have

$$t_m = \sum_{j=0}^{m} \left[\sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m-j-k} p^k q^{m-j-k} \right] s_j$$

and

$$\begin{aligned} t_{m+1} &= \sum_{j=0}^{m+1} \left[\sum_{k=0}^{m+1-j} \binom{n-j}{k} \binom{n-j}{m+1-j-k} p^k q^{m+1-j-k} \right] s_j \\ &= s_{m+1} + \sum_{j=0}^{m} \left[\sum_{k=0}^{m-j} \binom{n-j}{k} \binom{n-j}{m+1-j-k} p^k q^{m+1-j-k} \right] s_j \\ &\quad + \sum_{j=0}^{m} \binom{n-j}{m+1-j} p^{m+1-j} s_j. \end{aligned}$$

Therefore, we obtain

$$t_{m+1} - t_m = s_{m+1} + \sum_{j=0}^{m} \binom{n-j}{m+1-j} p^{m+1-j} s_j + \sum_{j=0}^{m} \sum_{k=0}^{m-j} \binom{n-j}{k} \frac{(n-j)! p^k q^{m-j-k} s_j}{(m-j-k+1)!(n-m+k)!} [(n-m+k)q - (m-j-k+1)].$$

1. By the assumption, we have $0 \le m \le \lfloor \frac{nq-1}{1+q} \rfloor$. Then $m \le \frac{nq-1}{1+q}$ and thus $m(q+1) \le nq-1$. On the other hand,

$$\begin{aligned}(n-m+k)q - (m-j-k+1) &= nq + k(1+q) - m(1+q) + j - 1 \\ &\ge nq + k(1+q) - (nq-1) + j - 1 \\ &\ge k(1+q) + j \ge 0.\end{aligned}$$

Hence $t_{m+1} \ge t_m$ for all $0 \le m \le \lfloor \frac{nq-1}{1+q} \rfloor$. Therefore,

$$t_0 \le t_1 \le \cdots \le t_{\lfloor \frac{nq-1}{1+q} \rfloor} \le t_{\lfloor \frac{nq-1}{1+q} \rfloor + 1}.$$

In particular, if $q \ge n$, then $\lfloor \frac{nq-1}{1+q} \rfloor + 1 = \lfloor \frac{n(q+1)-n-1}{1+q} \rfloor + 1 = n + 1 - \lceil \frac{n+1}{q+1} \rceil = n$. It follows that $t_0 \le t_1 \le \cdots \le t_n$. By the symmetric roles of p and q, we also obtain that $t_0 \le t_1 \le \cdots \le t_r \le t_{r+1}$, where $r = \max\{\lfloor \frac{np-1}{p+1} \rfloor, \lfloor \frac{nq-1}{q+1} \rfloor\}$. Moreover, $t_0 \le t_1 \le \cdots \le t_n$ whenever $\min\{p, q\} \ge n$.

2. Combining Lemma 4.1(2) and Proposition 4.3, we conclude that

$$t_s \ge t_{s+1} \ge \cdots \ge t_{2n}$$

for

$$s = \left\lceil \frac{n\left(G \circ (K_p \cup K_q)\right) - 1}{\min\left\{p, q, \frac{p+q-1}{2}\right\} + 2} \right\rceil = \left\lceil \frac{n(p+q+1) - 1}{\min\left\{p, q, \frac{p+q-1}{2}\right\} + 2} \right\rceil,$$

whenever $s \le 2n-1$, as claimed. □

In the case $p = q$, the results of Theorem 2 can be specified as follows.

Corollary 5.6 *Let $n = n(G)$ and $I(G \circ 2K_p; x) = \sum_{j=0}^{2n} t_m x^m$. Then the following assertions are true:*

1. $t_0 < t_1 < \cdots < t_n$;
2. $t_s \ge t_{s+1} \ge \cdots \ge t_{2n}$, *where* $s = \left\lceil \frac{n+2np-1}{p+\frac{3}{2}} \right\rceil$ *and* $s \le 2n-1$.

Proof 1. By Corollary 5.5, we have

$$t_m = \sum_{j=0}^{m} \binom{2n-2j}{m-j} p^{m-j} s_j, \quad \text{and}$$

$$t_{m+1} = \sum_{j=0}^{m+1} \binom{2n-2j}{m+1-j} p^{m+1-j} s_j = s_{m+1} + \sum_{j=0}^{m} \binom{2n-2j}{m+1-j} p^{m+1-j} s_j.$$

Hence,

$$t_{m+1} - t_m = s_{m+1} + \sum_{j=0}^{m} \frac{(2n-2j)! p^{m-j} s_j}{(m+1-j)!(2n-j-m)!} [(2n-j-m)p - (m+1-j)].$$

Since $p \geq 1$, it implies that

$$\begin{aligned}(2n-j-m)p-(m+1-j) &\geq (2n-j-m)-(m+1-j)\\ &= 2n-2m-1 \geq 0,\end{aligned}$$

for all $m \leq n-1$. Since $2n-2m-1$ is an odd number, then the condition $2n-2m-1 \geq 0$ is equivalent to the inequality $2n-2m-1 > 0$. Thus, $t_{m+1} > t_m$ for all $m \leq n-1$. Therefore, we obtain

$$t_0 < t_1 < \cdots < t_n.$$

2. Note that $n(G \circ 2K_p) = 2pn + n$ and $\alpha(G \circ 2K_p) = 2n$. By Theorem 2*(2)*, taking into account that $p = q$, we obtain

$$t_s \geq t_{s+1} \geq \cdots \geq t_{2n}$$

for

$$s = \left\lceil \frac{2np+n-1}{p+\frac{3}{2}} \right\rceil,$$

whenever $s \leq 2n-1$, as required. □

6 Conclusions

This paper explores the f-symmetric and unimodal properties inherent in the independence polynomials of graphs. More specifically, our results demonstrate that $I(G \circ (K_p \cup K_q); x)$ exhibits f-symmetry and unimodality for any graph G when p and q are sufficiently large.

A *very well-covered graph* G is defined as a well-covered graph without isolated vertices for which $n(G) = 2\alpha(G)$. For example, $G \circ K_1$ is always very well-covered for every graph G. In [6], it is demonstrated that every graph G can be viewed as an induced subgraph of a very well-covered graph whose independence polynomial is unimodal, using the k-th iterated leafy extension of G. However, the order of the graph resulting from the k-th leafy extension may be quite large. According to Theorem 1 and Corollary 6.2, it follows that each graph G can also be embedded as an induced subgraph of a non-well-covered graph $G \circ (K_p \cup K_q)$ with sufficiently large p and q, or even $p = q = 1$, such that its independence polynomial is unimodal. Moreover, since $\alpha(K_p \cup K_q) = 2$, it is worth mentioning that if $I(G; x)$ is real-rooted, then by [21, Theorem 5.4], $I(G \circ (K_p \cup K_q); x)$ is also real-rooted and, consequently, unimodal. Thus, Theorems 1 and 2, along with the results from [8] and [21], lead us to the following.

Conjecture 6.1 For any positive integers p and q, $I(G \circ (K_p \cup K_q); x)$ is unimodal for all graphs G.

By substituting $p = 1$ into Corollary 5.6*(1)* and applying the fact that the independence polynomial of $G \circ 2K_1$ is palindromic, as established in [31], we obtain the following.

Corollary 6.2 ([26, Theorem 2.2]) *$I(G \circ 2K_1; x)$ is unimodal with the mode at n.*

In the case where G is a tree of order n, $G \circ 2K_1$ is a tree of order $3n$. Hence, G can be considered as a sub-tree of a larger tree whose independence polynomial is unimodal, regardless of whether $I(G; x)$ is unimodal. It motivates the following.

Conjecture 6.3 Let T' be a sub-tree of a tree T. If $I(T; x)$ is unimodal, then $I(T'; x)$ is unimodal as well.

Clearly, Conjecture 1.1 is equivalent to Conjecture 6.3.

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Declarations

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